Contract 80GSFC18C0120

NASA
GODDARD SPACE FLIGHT CENTER

STATEMENT OF WORK

FOR

ELECTRICAL SYSTEMS ENGINEERING SERVICES III (ESES III)

FOR THE

APPLIED ENGINEERING AND TECHNOLOGY DIRECTORATE (AETD)

6 November, 2017

Table of Contents

INTRODUCTION

The National Aeronautics and Space Administration (NASA) was established to plan, direct, and conduct aeronautical and space activities for peaceful purposes for the benefit of all mankind. The operational aspects of NASA's work are divided among field installations around the country and involve research and development activities under the responsibility of the technical program offices at NASA Headquarters.

The Goddard Space Flight Center (GSFC) - located in Greenbelt, Maryland, along with its Wallops Flight Facility in Virginia, supports NASA's Science Mission Directorate (SMD), as well as the other NASA mission directorates, and the Office of Chief Technologist. The GSFC is chartered to expand the knowledge of the Earth and its environment, the Solar System, and the Universe through observations from space. To this end, the GSFC's primary emphasis is in scientific investigation, in the development and operation of space systems, and in the advancement of essential technologies. In accomplishing this responsibility, the GSFC has undertaken a broad program of scientific research, both theoretical and experimental, in the study of space phenomena and earth sciences. The program ranges from basic research to flight experiment development, and from mission operations to data analysis.

Within the GSFC, the Applied Engineering and Technology Directorate (AETD) plans, organizes, and conducts a broad range of technical research and development activities in support of science applications. The AETD is responsible for providing engineering expertise and support in the design, development, fabrication, integration, test, and verification of components, subsystems, systems, science instruments, and complete spacecraft for multiple projects. The specific components, subsystems, systems, and science instruments are ultimately integrated into the spacecraft to form a science observatory. It is these observatories that are launched to fulfill the mission of the GSFC. The AETD comprises five engineering divisions: the Mechanical Systems Division (MSD), the Software Engineering Division (SED), the Instrument Systems and Technology Division (ISTD), the Electrical Engineering Division (EED), and the Mission Engineering and Systems Analysis Division (MESA).

To fulfill these responsibilities and ultimately achieve their missions, the AETD must acquire a wide range of engineering services in support of its divisions to implement the GSFC mission.

SCOPE OF WORK

The principal purpose of this contract is to provide electrical/electronic engineering support services and related work to EED, ISTD, SED, MESA, related organizations and other NASA Centers and locations specified in task orders, for the study, design, development, fabrication, integration, testing, verification, and operations of space flight, airborne, and ground system hardware and software, including development and validation of new technologies to enable future space and science missions.

To this end, the Contractor shall provide on/off-site multidisciplinary engineering services, pursuant to task orders issued by the Contracting Officer. These services shall include the personnel, facilities, and materials (unless otherwise provided by the Government) to accomplish the tasks.

Task orders will be issued to perform services in the following areas: the design, development, analysis, fab/build, assembly, integration, test, verification, and launch and mission support operations of parts, components, subsystems, systems, science instruments/experiments, airborne/suborbital systems (e.g., aircraft, sounding rockets, UAVs, balloons), free-flying spacecraft, airborne/suborbital/orbital payloads, and Space Station payloads. To include related ground support equipment, simulators, non-flight models, and prototypes. Additionally perform candidate, feasibility, and systems definition studies; project management; systems engineering; analysis; preliminary design; detailed design; fabrication; assembly; integration; test and verification; test instrumentation; data systems management; launch and post-launch operations; research and technology unique to system development; parts screening and materials selection/analysis; documentation; maintenance; sustaining engineering; configuration management; performance assurance; systems safety; and contamination control.

I. GENERAL RESPONSIBILITIES

The Contractor's responsibilities shall include the management of personnel, timely and effective implementation of task orders, control and monitoring of contract and subcontract performance, management of scheduled deliveries, and timely and effective reporting to the Government. These responsibilities shall also include efficient cost management methods as well as procedures to ensure that the Government is aware of task order status and progress achieved.

The Contractor shall be responsible for ensuring that all contractor and subcontractor personnel engaged in performance of this Statement of Work have appropriate qualifications, knowledge, and certification to perform work in accordance with the task orders. Most importantly, to ensure safety of personnel and property, all contractor personnel shall be familiar with and adhere to the Center's Occupational Safety Program (GPR-1700.1B) and AETD's Safety Manual (500-PG-8715.1.2C) as well as specific division, and facility/lab safety policies (ref GDMS, GIDs).

II. PERFORMANCE MEASUREMENT

Performance-based statements of work/specifications will be used for establishing contract requirements. Therefore, each task order issued by the Contracting Officer will include, as a minimum, the following:

1. Statement of Work, including the requirements to be met, the standard(s) of performance/quality of work, and required deliverables (or other output)

2. Performance Specification (if applicable)

3. Applicable Documents (if required)

4. Period of Performance

5. Surveillance Plan

The Contractor shall be required to adhere to the performance measurements detailed in each task order.

III. TASKS

Services shall be required in one or more of the areas described in the scope above for any given task order. Services within the scope of this Statement of Work and specified in task orders shall include, but not be limited to, the specific services delineated in the following sections.

FUNCTION 1 – PRE-FORMULATION AND FORMULATION SERVICES: CANDIDATE, PRELIMINARY ANALYSIS, AND SYSTEMS DEFINITION STUDIES

The Contractor shall provide study services with emphasis on electrical systems engineering that integrate the aspects of flight systems, ground systems, instrument systems, and launch systems.

In general, the Contractor shall:

- Produce pre-formulation and formulation phase study inputs for spacecraft, airborne/suborbital craft, instruments and ground systems.

- Develop mission needs (mission objectives, measurement concept, and instrument concept) and mission design (mission requirements, architectural design, and operations concept).

- Develop preliminary, relative cost and schedule estimates based on design alternatives, and identify and assess high-risk elements in designs.

- Document the history of design, qualification, flight experience, and modifications where existing components or subsystems are to be utilized.
- Identify interface requirements for pre-launch, launch, on-orbit servicing, or retrieval of flight hardware.
- Define interface engineering and management requirements.

- Prepare mission systems and operations documentation.

- Prepare requirements and specification packages that conform to applicable standards defined within Task Order statement.

- Identify interfaces and prepare interface control documents.

- Provide technical inputs for problem-solving and/or design inputs in selected spacecraft, instruments, suborbital craft, ground system, and data disciplines.

- Analyze various reports (i.e., progress reports) delivered by the GSFC mission Contractor(s) and provide recommendations to the project.

- Provide liaison and coordination services for project activities.

- Perform hardware and software design and trade-off analyses.

- Provide design services that include performance of preliminary design (leading to a Preliminary Design Review) of the subsystems, components, and assemblies that comprise the instrument/spacecraft/platform/launch system.

A. Candidate Study Services

The Contractor shall provide study services for the conceptual development of subsystems and systems thereby participating in the identification of scientific objectives, mission requirements and technical concepts. Study products produced during this phase shall include, but not be limited to:

- Strategic technology planning;
- Integration of joint missions, partnerships, and other collaborative efforts;
- Research/science/technology/cost trade studies;
- Candidate operations concepts;
- Candidate system architectures;
- Cost, schedule, and risk estimates;
- Research and technology unique to system development;
- Customer development support and outreach; and
- Mission justification and objectives.

B. Electrical and Electronics Systems Preliminary Analysis Study Services

The Contractor shall provide preliminary Electrical and Electronics Systems analysis study services focusing on analyzing mission requirements and establishing mission architectures in order to demonstrate that a credible, feasible design(s) exist(s). The Contractor shall develop top-level requirements and evaluation criteria, identify alternative operations/logistics concepts, and identify project constraints and system boundaries. Contactor activities shall include the consideration of alternative Electrical and Electronics Systems design concepts including feasibility and risk studies, cost and schedule estimates, and advanced technology requirements. The Contractor shall prepare for and support the appropriate Phase A project and technical reviews and prepare Phase A project documentation as appropriate (ref the NASA Systems

Engineering Handbook, SP-6105 and NPR 7120.5, "NASA Space Flight Program and Project Management Requirements").

C. Electrical and Electronics Systems Definition Study Services

The Contractor shall provide Electrical and Electronics Systems definition and preliminary design study services to establish (and evolve) project baseline(s) which includes a formal flowdown of project-level performance requirements to a complete set of system and subsystem design specifications for both flight and ground elements. The Contractor shall produce technical requirements that are sufficiently detailed to establish firm schedules and cost estimates for the project. Phase B baseline information shall be developed including Electrical and Electronics Systems requirements and verification requirements matrices, system architecture and work breakdown structures, operations concepts, "design-to" specifications at all levels, and project plans including schedule, resources, acquisition strategies, and risk management. The Contractor shall prepare for and support the appropriate Phase B project and technical reviews and prepare Phase B project documentation as appropriate (ref the NASA Systems Engineering Handbook, SP-6105 and NPR 7120.5, "NASA Space Flight Program and Project Management Requirements).

FUNCTION 2 – IMPLEMENTATION PHASE SERVICES

The Contractor shall provide services to design, develop, fabricate/code, assemble, unit test, system integrate, verify, integrate, deploy, and operate hardware and software on spacecraft, platform, and/or payload as defined by this Statement of Work. The implementation phase services shall include:

A. Project Management

The Contractor shall provide management services, including establishment of a management organization that ensures that all the assigned task objectives are accomplished within specified schedule and cost constraints. Management shall provide frequent and timely status to the Government via cost, schedule, progress and other reports during all phases of work.

B. Electrical and Electronics Systems Engineering

The Contractor shall provide systems engineering support for project development, reporting progress and conformance to appropriate practices and specifications (per GPR 7123.1 Systems Engineering). The Contractor shall support system engineering activities required during the Implementation phases including design (Phase C), development (Phase D), and operations (Phase E).

The Contractor shall support Phase C Design systems engineering activities to complete the detailed design of systems and associated subsystems, and operational systems, ensuring compliance with GSFC's Gold Rules (GPR 8070.4 and GSFC-STD-1000). The Contractor shall add remaining lower-level design specifications to the systems architecture, refine requirements document and verification plans, prepare interface documents, evolve and mature the design, augment baseline documentation, perform and archive trade studies, and complete manufacturing plans. The Contractor shall prepare for and support the appropriate Phase C project and technical reviews and prepare Phase C project documentation as appropriate (ref the NASA Systems Engineering Handbook, SP-6105 and NPR 7120.5).

The Contractor shall support Phase D Development systems engineering activities to build the subsystems (including operations systems) and integrate them to create the system(s). The Contractor shall develop verification procedures, archive documentation for verifications performed, audit "as-built" configurations, document lessons learned, prepare operator's manuals and maintenance manuals, and perform operational verification. The Contractor shall prepare for and support the appropriate Phase D project and technical reviews and prepare Phase D project documentation as appropriate (ref the NASA Systems Engineering Handbook, SP-6105 and NPR 7120.5).

The Contractor shall support Phase E Operations systems engineering activities to perform the mission and then dispose of the system in a responsible manner. The Contractor shall prepare for and support the appropriate Phase E project and technical reviews and prepare Phase E project documentation as appropriate (ref the NASA Systems Engineering Handbook, SP-6105 and NPR 7120.5).

Specific tasks over the various phases shall include:

1. Instrument Avionics and Electrical Systems Engineering
   The Contractor shall perform instrument avionics and electrical systems engineering functions that include, but are limited to:

   a. Instrument Data Processing Development & Support ;
   b. Instrument Architecture & Design Development;
   c. Requirements Analysis, Identification and Management;
   d. Verification and Validation;
   e. Interface and Interface Control Documents;
   f. Instrument Environment;
   g. Technical Resource Budget Tracking;
   h. Risk Analysis, Reduction and Management
   i. Instrument System Milestone Review;
   j. CM; and
   k. Instrument Systems Engineering Management Plan.

C. Multidisciplinary Analyses Services

The Contractor shall provide analytical and detailed design support that includes multi-disciplinary system analyses and trade studies involving, but limited to, mechanical, thermal, optics (including radiometrics and stray light), contamination, control, guidance, navigation, detector, electrical, electronic, and software aspects of flight and ground systems, including associated ground support equipment. Services shall include the definition, development and use of models and simulations to study and quantify system performance and to conduct system trade studies. This includes, but is not limited to, such specific efforts as electromagnetic interference/electromagnetic compatibility (EMI/EMC) analysis, environmental testing, magnetics testing, thermal vacuum testing, systems performance error budgets, power and weight budgeting and tracking, microphonics analyses, fracture control analyses, microwave communication system analyses (including link margin availability), controls-structures analyses, control-structural-thermal-optical analyses, instrument system analyses (including system sensitivities), computer performance analyses (including Central Processing Unit (CPU), memory, simulations, etc), and systems reviews of selected critical subsystems. Services shall also include the development of measurement tools and models, analysis of measurement data, defect tracking, process improvement, cost estimation of hardware and software systems, modeling of organizational processes, electrical parts analyses, electrostatic discharge prevention, printed wiring board layouts, fiber optic cable construction (test, validation, & assembly), electro-optic/photonic analyses, thermal analysis, printed-wiring board fabrication, mechanical enclosure design, and technical writing documentation.

D. Detailed Design Services

The Contractor shall provide design services that include performance of detailed (leading to a Critical Design Review) design of the subsystems, components and assemblies that comprise the instrument/spacecraft/platform in accordance with current NASA/GSFC design standards, specifications and best practices or commercial equivalent that meets/exceeds their intent. This effort includes hardware and software (flight and ground) as well as ground support equipment (electrical, thermal, contamination, mechanical, and cryogenic). Documentation, including technical reports, drawings, schematics, block diagrams, layouts, parts and materials list, and equipment lists, shall be provided. Specific tasks shall include:

1. **Optical Specific Tasks** – The Contractor shall provide optical design services, including the design, development and analysis of the following:

    a. Geometric optical design;
    b. Diffraction analysis;
    c. Tolerancing of components;
    d. Gaussian beam analysis;
    e. Interferometry;
    f. Radiometry;
    g. Stray light/energy analysis;
    h. Alignment and calibration; and
    i. Optical communications systems design.

2. **Mechanical Specific Tasks supporting Electromechanical design** – The Contractor shall provide mechanical design services, including the design, development and analysis of the following:

    a. Assembly and subassembly hardware to house flight components;
    b. Mechanisms and electromechanical assemblies;

c. Component mount design;
d. Structure;
e. Ground support and bench test equipment;
f. Analytical studies, including structural analyses (stress, dynamics, thermal, deformation, jitter, etc.); mechanical alignment; interfaces; fracture control; controls analysis; mechanism dynamics; composite material/structure dimensional stability; and load analyses;
g. Associated instrumentation and control systems;
h. Finite element model analysis;
i. Flight loads analysis;
j. On-orbit loads analysis;
k. Stress analysis;
l. Dynamic analysis;
m. Mechanical design and drawing production/verification.

3. **Electromechanical -Specific Tasks** – The Contractor shall provide electromechanical design services, including the design, development and analysis of the following:

    a. Concept trades, concept design, and detailed design of electromechanical systems and their components;
    b. Fabrication, assembly, and testing services, including life testing;
    c. Magnetic bearings, active/smart structures, vibration isolation, and large aperture, lightweight systems;
    d. Analyses using Computer-aided Design (CAD) and simulation tools utilizing hardware and software compatible with those used by the Mechanical Systems Division (MSD);
    e. Set-up and operate electronic design and test equipment compatible with equipment used by the MSD;
    f. Design, analysis, selection, implementation, and testing of bearings, flex-pivots, and flexures;
    g. Robotics;
    h. micro electromechanical systems (MEMS);
    i. Micro-miniaturization of mechanisms, or sub-assemblies of conventional mechanisms;
    j. Small scale structures, materials, fabrication techniques, packaging, photolithography, lithography electroforming and molding (LIGA in German), interfacing with macro-components, micro-actuators, focused ion-beam milling and welding, and coatings and tribology issues; and
    k. Single Input Single Output (SISO) and Multiple-Input-Multiple-Output (MIMO) controllers.

4. **Thermal Specific Tasks supporting Electromechanical design** – The Contractor shall provide thermal design services, including the design, development and analysis of the following:
    a. Thermal system design, active and passive;
    b. Component/subassembly/assembly cooling, including electronics;
    c. Associated instrumentation and control systems;
    d. Thermal laboratory support; and

e. Thermal vacuum test support.

5. **Detector Specific Tasks** – The Contractor shall provide engineering services for state-of-the-art detection systems requiring low noise levels and calibrations traceable to physical standards including the design, development, test, and analysis of the following systems:
   a. Radio Frequency (RF);
   b. Submillimeter wave;
   c. Microwave;
   d. Millimeter wave;
   e. Infrared;
   f. Visible;
   g. X-ray;
   h. Gamma-ray;
   i. Neutral and charged particle detection;
   j. MEMS; and
   k. Nanotechnology.

6. **Electrical/Electronic Specific Tasks** – The Contractor shall provide electrical/electronic design services, including the design, development and analysis of the following:
   a. Flight electronic subsystems;
   b. Command and data handling systems;
   c. Flight and ground data systems;
   d. Low noise electronics;
   e. Digital and microprocessor based designs;
   f. Analog control circuits;
   g. Control systems;
   h. Low voltage systems;
   i. High voltage systems;
   j. Power supplies with programmable voltage and current outputs;
   k. Electromagnetic field analysis;
   l. Electromagnetic compatibility and interference (EMC/EMI);
   m. Test circuitry and equipment, including interface to computer-based systems;
   n. Bench test equipment;
   o. Electrical ground support equipment;
   p. High speed electronics;
   q. Communication electronics;
   r. Microcontroller based systems and embedded systems;
   s. Flight and test harnesses (conventional wire and fiber optic);
   t. Breakout boxes;
   u. Pyrotechnics;
   v. Simulators;
   w. Field Programmable Gate Array (FPGA)-based reconfigurable computing systems;
   x. Analog and mixed-signal Application-Specific Integrated Circuit (ASIC) devices; and
   y. Programmable logic devices (FPGA and ASIC devices).

In addition, analyses include, but are not limited to: Worst-case analysis, Parts Stress analysis, reliability analysis, thermal analysis, structural analysis, signal integrity analysis and performance/throughput analysis. Peer design reviews shall be an integral part of the overall development cycle. Test support includes shift support for environmental tests (thermal-vacuum, EMI, vibration, etc).

7. Ground Software and Systems for post launch sustaining engineering

   The Contractor shall perform new or ongoing development and sustaining engineering of ground software and data systems. Contractor activities may include participation as a part of a mission's or project's ground system development team. Ground systems development includes software design, development, test, and deployment, and associated hardware integration, test, and delivery. The software may include commercial off-the-shelf (COTS), government off-the-shelf (GOTS), new development, or some combination of the aforementioned. The Contractor may perform hardware systems and associated material procurement/inventory management. The procured hardware systems shall comply with established NASA safety protocols including at least Hazardous Material Management and Electrostatic Discharge (ESD).

   Ground Software Services shall include:

   a. Ground Software Architectures;
   b. Ground Software Development; and
   c. Ground Software Sustaining Engineering.

8. Flight Software System Verification, and Validation

The Contractor shall support flight mission operations including build verification testing, system validation testing, and flight software and flight operations test tools development for the mission.

Validation and Verification Functional Areas**:**

a. Utilize flight software test-beds and simulators required for verification and validation of flight and ground software related products;
b. Support mission readiness testing to include communication and functional test of the mission ground system, ground station interfaces, and end-to-end testing of the communication path from the spacecraft to the ground and/or science operations systems;
c. Generate and/or review Verification Plans;
d. Support the design, drawing, and specification reviews;
e. Prepare documentation and/or review of system qualification requirements;
f. Prepare and/or review hardware and software integration plans and procedures, and witnessing execution;
g. Prepare and/or review detailed functional and environmental test plans and procedures, and witness test execution;

h. Prepare and/or review plans for launch site checkout, integration and testing of flight systems, including adequacy of the launch site facility;
i. Analyze data from spacecraft telemetry data sources to ensure total system compatibility; and
j. Analyze flight performance from flight data.

9. Data System Development

The Contractor shall provide data system engineering, and software engineering services to design, develop and deploy science systems and applications for science operations, data processing, data management, and data analysis and visualization. Data system engineering may include problem definition, solution analysis, process planning and control, system documentation development and maintenance, system integration, and product evaluation. Software system engineering may include requirements definition and analysis, software design, process planning and control, software documentation development and maintenance, verification, validation and test, and software integration. Software engineering may include detailed software design, implementation, and unit testing. The data system software may include COTS, GOTS, new development, legacy code, or some combination of the aforementioned. Data systems development may include hardware definition, integration, test and deployment. The Contractor may provide science operations support for ongoing missions.

Software Science System Development Services shall include:

a. Data Systems Engineering;
b. Data Operations Systems;
c. Data Processing Systems;
d. Data Management Systems; and
e. Data Analysis and Visualization Applications.

10. **Power Specific Task** – The Contractor shall provide power design services, including the design, development, and analysis:

a. Power system design and analysis tools for energy balance and regulation;
b. Spacecraft power management and distribution electronics;
c. Photovoltaic energy conversion cells, arrays, and associated ground testing;
d. Electrochemical energy storage cells, batteries, and associated destructive physical analysis and ground testing; and
e. Payload and instrument low and high voltage power conditioning electronics (converters, filters, regulators).

11. **Radiation Effects and Analysis (REA) Specific Tasks** – The Contractor shall provide radiation testing and analysis services for the Electrical, Electronic, and Electromechanical (EEE) and photonic components, including:

a. Space radiation environmental analysis and specifications including nuclear interaction simulations;
b. Design and development of test plans and test suite hardware and software to

support research and flight project efforts;

c. Documentation of radiation test techniques and research results;
d. Determination of mission-specific system level impact of radiation test results and evaluate mission radiation risk assessment;
e. Evaluations of parts list for radiation vulnerable devices;
f. General REA services in the area of graphics, schedules, and reporting requirements;
g. Development of user interface software for REA-developed environment models;
h. Operation and maintenance of GSFC's radiation facility and Instrument calibration;
i. Design and fabrication of radiation flight experiments and providing services to support data analysis;
j. Analysis of in-flight data on experimental and operational systems to determine system performance in radiation environment;
k. Analysis of instrument data to support space environment modeling;
l. Analysis and performance of tests at established radiation test facilities (requires travel);
m. Dissemination of radiation effects research results via paper and presentations;
n. Database development and management of the REA radiation effects test data; and
o. Curator capabilities for maintenance and upgrade of the Radiation and Analysis (REA) WEB site.

12. **Component Technologies Specific Tasks** – The Contractor shall provide component technology specific services for design, development, and analysis, including:

    a. Non-destructive Evaluation (NDE) of complex integrated circuits employing Finite Element Analysis, Scanning Acoustic Microscope, X-Ray, and thermal analyses;
    b. Analysis of novel microelectronic materials;
    c. Environmental analysis of active and passive fiber optic components;
    d. Microelectronic assembly, including ASICs and custom hybrids; and
    e. Lightweight, low-power transmitters and receivers for active and passive microwave instrument system.

13. **Environmental Testing Support for Specific Tasks** – The Contractor shall provide environmental test support for the Avionics and Electrical Systems, including:

    a. Vibration;
    b. Acoustic;
    c. EMI/EMC;
    d. Thermal; and
    e. Vacuum.

14. **RF Specific Tasks** – The Contractor shall provide RF design services, including the design, development and analysis of the following:

    a. RF Component design;
    b. Communications systems design;
    c. Communications systems component design;

d. RF Modulator design (Binary Phase-shift keying (BPSK), Quadrature Phase-shift keying (QPSK), 64 Quadrature Amplitude Modulation (QAM), Code Division Multiple Access (CDMA), 8 Phase-shift keying (PSK) etc.);
e. Antenna Design L - Ka bands;
f. Phased Array antenna design;
g. Solid State Power Amplifier Design S to Ka bands;
h. Transponder design S to Ka bands; and
i. Millimeter wave systems instrument and components design (50 –300 GHz).

In addition, the Contractor shall provide independent systems engineering analysis and evaluation in support of customer mission-related activities as well as analysis and support of new capabilities considered for inclusion into the NASA Space Network (SN) and Ground Network (GN) as well as Global Positioning Satellite (GPS) Navigation. This also includes RF communications systems engineering support to independently evaluate and assess current space and ground network capabilities, system improvements, and new services to meet future mission needs. Additionally, this effort may include performing SN analyses for current space elements (TDRS Flight-1 through Flight-10 and TDRS K, L , M & N) including the ability for modeling of multiple access interference signals and the corresponding current ground elements at the White Sands Complex (WSC), Guam Remote Ground Terminal, and Space Network Expansion (SNE) Terminals. This effort may encompass analysis related to the Ground Network assets that are used for mission support to include analysis for other networks and commercial ground stations. Maintenance & development of analytical tools to perform the required analysis and associated data bases to support the analysis are included in this effort. This effort supports standards and protocol activities. Maintenance & extension of Space and Ground Network Users Guides are also encompassed in this effort. This effort includes the development and review of RF Interface Control Documents as well as technical reports on special studies. Analysis of new modulation schemes and transmission techniques including lasers are additionally covered under this activity including the ability to perform analysis of atmospheric effects of higher communication frequency bands (e.g. 37 GHz). Ability to perform dynamic analyses for Spacecraft (S/C) & launch vehicles which requires expert knowledge of trajectories, orbital mechanics & the 3-dimension modeling S/C to enable S/C & Expendable Launch Vehicle (ELV) antenna design to be compatible with the SN & GN.

Qualified personnel with security clearances of at least a DoD SECRET level (as defined in the specific Task Order), may be required by the Contractor to support certain efforts. The Contractor must also comply with applicable NASA, DoD, National Industrial Security Program Operating Manual (NISPOM) and Director of

Central Intelligence Directives (DCIDs) security regulations.

15. **<u>GN&C Component & Hardware Specific</u>** <u>Tasks</u> – The Contractor shall perform specific GN&C component and hardware systems engineering tasks, including:

1. Component and Hardware Systems Engineering;
2. Test Bed & Simulator Design/ Development Services;

E. Fabrication, Assembly and Testing Services

The Contractor shall provide flight (including protoflight), and non-flight (including prototype) hardware fabrication and assembly facilities, and support for mechanical subassemblies, components, electronic assemblies, electromechanical devices, thermal control devices and subsystems, and thermal flight experiments. All fabrication and assembly support shall be in accordance with the workmanship requirements of NASA-STD-8739.3 thru 8739.5 and Institute for Printed Circuits (IPC) 6011, 6012, 6013, 6015, 6016, and 6018, as well as all subsequent updates to these documents. All fasteners used in assembling or installation shall conform to Goddard Procedures and Guidelines (GPG) 541-PG-8072.1.2, GSFC fastener integrity plan. Also provide support to fabricate mechanical ground support equipment, special test and evaluation equipment (including electronic equipment) necessary to support the operation of all mechanical hardware. In situations where hardware fabrication is required in a quick reaction mode and the Contractor decides to perform the task under subcontract, the Contractor shall minimize both the subcontract implementation and fabrication phases of the task. Subcontractors used for fabrication and/or assembly shall be AS9001C compliant.

The Contractor shall provide fabrication, assembly and testing services, including breadboards, engineering models, protoflight models, and flight models at all levels of assembly specified by this Statement of Work, including:

1. **Planning Specific Tasks** – The Contractor shall provide planning services, including:

    a. Implementation and maintenance of overall production and quality engineering plans; and
    b. Manufacturing, integration and test plans, describing sequences, qualification and acceptance test levels, and facilities needed to accomplish assembly, integration, alignment, testing, quality control, and checkout.

2. **Fabrication Specific Tasks** – The Contractor shall provide fabrication services, including:
    a. Optical, mechanical, detector, electrical/electronics, and microwave, including antennas;
    b. Ground support equipment, including mechanical and electrical, and optical;
    c. Laboratory control systems;
    d. Wiring harnesses;
    e. Special parts;
    f. Surface mount printed circuit boards, including leadless chip carriers and chip-on- board techniques; and
    g. Composites.

3. **Assembly Specific Tasks** – The Contractor shall provide assembly services, including:

   a. Optical, mechanical, detector, electrical/electronics, and microwave, including antennas;
   b. Ground support equipment including mechanical, electrical, and optical
   c. Test equipment and fixtures;
   d. Wiring harnesses;
   e. Support for the operation of GSFC's Board Layout Facility in Building 5 and Assembly Facility in Building 35;
   f. Active thermal control devices (heaters, thermostats, thermocouples, thermisters, heat pipes, Cold Plates, etc.); and
   g. Hardware protective coatings.

4. **Logistics Specific Tasks** – The Contractor shall provide logistics services, including:

   a. Identification of critical spares and material;
   b. Storage and control of critical spares and material; and
   c. Shipment of materials, supplies, subsystems, ground support equipment, systems, and flight systems to and from integration, test and launch facilities.

5. **Testing Specific Tasks** – The Contractor shall test and/or participate in the GSFC's testing and qualification of hardware and software, including retesting/requalification of spare units and breadboards previously developed for flight projects. These tests shall be conducted in accordance with Government-approved procedures and shall include both functional and environmental tests in accordance with GSFC-STD-7000 "General Environmental Verification Standard (GEVS). Functional tests shall be designed and performed to demonstrate compliance with the operating requirements of the system. Environmental tests shall be designed and performed using environmental conditions that meet the launch, safety, and operations requirements of the assigned task. The Contractor shall perform the following:

   a. In-process testing during the fabrication process to demonstrate that the design meets the requirements specified. In-process testing shall include:

      1. Component value measurements and verification of polarity prior to installation and after installation, where feasible;
      2. Resistance checks of point-to-point wiring and cross tie points, where applicable;
      3. Hi-pot operations from component-to-component, component-to-frame, etc. in accordance with the applicable GSFC specification or procedure, as required by and specified in a task order;
      4. Leak/pressure testing at the lowest level of assembly possible and throughout the assembly stages;
      5. X-ray, dye penetrant, and eddy current inspections, as well as other forms of nondestructive analysis; and
      6. Tests to develop/validate models for structural, mechanical, thermal, optical, and electronic components and assemblies.

b. Functional testing, including:

    1. Verification of operational characteristics of components and equipment;
    2. Testing at Government facilities; and
    3. Testing and documentation to verify accuracy, repeatability, and stability while operating under simulated flight conditions.

c. Flight qualification testing on units that have successfully completed functional tests and have been prepared for space flight. These tests may be conducted at any of the levels of assembly specified in this Statement of Work, including on the spacecraft. The qualification tests shall be carried out in a test environment specified by the task order. The Government may provide test facilities and/or test equipment to the Contractor, as specified in the task order. Flight qualification testing shall include:

    1. Vibration/Shock;
    2. Magnetic;
    3. Thermal vacuum;
    4. Thermal balance;
    5. Static loads;
    6. Acoustics;
    7. Mass properties;
    8. Alignment;
    9. Electromagnetic interference (EMI);
    10. Electromagnetic compatibility (EMC);
    11. Gravity effects;
    12. Radiation effects;
    13. Modal survey;
    14. Optical performance and characterization;
    15. Deployments; and
    16. Mechanism performance.

F. Integration, Test, and Verification Services

The Contractor shall provide engineering and test-conductor services that include integrating and verifying the flight, ground systems, and science data system/applications in accordance with applicable documentation and specifications, preparing test procedures, documenting all nonconformances and dispositions, calibrating the system and its ground support equipment, and providing operating manuals, reference documents, training, and launch site support.

1. **Integration, Test and Verification Specific Tasks** – The Contractor shall provide integration, test, and verification services, including:

    a. Major program reviews;
    b. Space flight subsystems;
    c. Space flight instruments;
    d. Space flight payloads;
    e. Airborne/Suborbital craft instruments;
    f. Ground instrumentation;
    g. Ground support systems;
    h. Science data systems/applications;
    i. Spacecraft and science operations control rooms; and
    j. Airborne/Suborbital craft subsystems.

Integration and test services may need to be supported at various locations, including vendor sites, NASA Centers, and Military sites.

G. Laboratory and Test Instrumentation Services

The Contractor shall provide the services necessary for conceptualization, prototyping, system engineering, design, development, integration, test, sustaining engineering, maintenance and utilization of laboratory and test instrumentation.

1. **Laboratory and Test Instrumentation Specific Tasks** – The Contractor shall provide laboratory and test instrumentation services, including:

    a. Optics/Fiber optics;
    b. Detector engineering;
    c. Laser communication and ranging;
    d. Electrical ground support equipment;
    e. Subsystem bench test equipment;
    f. Special instrumentation;
    g. Thermal/Cryogenics;
    h. Contamination;
    i. Operating power test facilities at the GSFC
        • Large Area Pulse Solar Simulator;
        • Battery handling & conditioning laboratory; and

- High voltage partial discharge laboratory.

j. Calibration;
k. Crosstalk;
l. Microphonics analyses;
m. Analysis of instrument and spacecraft subsystem interaction; and
n. RF communications engineering and millimeter wave services
   - Operating microwave test equipment, including cryogenic testbeds;
   - Operating RF equipment, including radiating sources; and
   - Supporting antennae test facilities, including anechoic chamber.

H. Data Systems Management Services

The Contractor shall provide data systems management services, including:

a. Developing, reviewing, and analyzing software requirements and specifications;
b. Contributing to the design, development, validation, implementation, certification, and maintenance of ground or on-board computer system simulators/emulators, including validation of flight systems software for ascent, transfer, or on-orbit phases and near real-time reprogramming and validation of modifications for recovery from anomalous situations; and
c. Analyzing the design and implementation of simulators/emulators for ground crew training, systems testing and procedure validation.

I. Launch and Post-Launch Operations Services

The Contractor shall supply launch and post-launch services, including:

1. **Launch Site Preparation Specific Tasks** – The Contractor shall provide subsystem services at the launch site, including:

   a. Payload system and its support equipment;
   b. Interfaces to the mission operations control centers;
   c. Technical services to facilitate interfacing with the launch site organization;
   d. Development of launch site support requirements; and
   e. Providing support in shipment of the flight hardware and associated support equipment to and from the launch site.

2. **Launch Operations Specific Tasks** – The Contractor shall provide launch operations services, including:

   a. Assuring flight readiness of the delivered subsystem;
   b. Pre-launch testing of the delivered subsystem;
   c. Operation of associated ground support equipment; and

d. Services to the launch vehicle team for payload integration to the vehicle at the launch facility.

3. **Mission Operation Support Specific Tasks** – The Contractor shall provide mission operation services, including services for the payload and for carrier and flight support system during mission operations.

4. **Landing and De-Integration Specific Tasks** – The Contractor shall provide landing and de-integration services, including services at the landing site for payload de-integration, post-flight testing, and payload shipment. This shall include airborne/suborbital craft and payloads recovery.

5. **Refurbishment of Recovered Systems Specific Tasks** – The Contractor shall provide refurbishment services for recovered flight systems.

6. **Data Reduction Specific Tasks** – The Contractor shall provide data reduction services and shall compile and analyzing systems performance data during and after the mission.

7. **Documentation Specific Tasks** – The Contractor shall provide post-flight summary reports, analyzing the performance of the subsystem during flight.

J. Mission Assurance and System Safety Services supporting Integration and Test activities including launch site

For all levels of flight hardware and software provided by the Contractor and specified by this Statement of Work, the Contractor shall establish and maintain a mission assurance program commensurate with mission requirements as specified by the task. The mission assurance program shall incorporate a system safety program which meets the requirements of NASA-STD-8719.24 (Base), "NASA Expendable Launch Vehicle Payload Safety Requirements" and NASA-STD-8719.24 (Annex), "Annex to NASA-STD-8719.24 NASA Expendable Launch Vehicle Payload Safety Requirements", and NSTS 1700.7B (Addendum) "Safety Policy and Requirements for Payloads Using the International Space Station".
The Contractor shall establish and maintain practices, procedures, and processes that are AS9100C, "Quality Management System Standard", or ISO 9001 compliant when applicable.

1. **Performance Assurance Specific Tasks** – The Contractor shall provide performance assurance services, including:

a. Reviewing payload designs to assure their compliance with performance assurance, reliability, and safety specifications;
b. Developing, analyzing, and monitoring performance assurance, reliability,

system safety plans, and procedures, fabrication assembly, integration and test, verification, and launch support; and

c. Analyzing basic plans for system safety, contamination control, integration, and testing of subsystems and systems.

K. Configuration Management Services

The Contractor shall provide overall management and oversight of the Configuration Management (CM), Documentation Management (DM), and Quality Control Management (QCM) disciplines throughout the life cycle of flight hardware and software provided within the scope of this Statement of Work. Each discipline shall require the development, establishment, and implementation of procedures and processes and establishment of mechanisms and tools for consistency.

The Contractor shall support the planning, identification of processes, and leading GSFC Project efforts in these disciplines. This support shall also include the necessary planning and associated process development for the GSFC Project in meeting conformance requirements to NASA procedures and guidelines as well as the ISO standards.

The main CM/DM/QCM functions shall include:

- Configuration identification, configuration control, configuration accounting and reporting;
- Configuration verification and configuration auditing; and
- Implementation and maintenance of a DM system.

The Contractor shall be responsible for providing the necessary tools and databases to accomplish the above functions; developing and establishing procedures and guidelines and training in the configuration management, documentation management, and ISO 9001 disciplines.

L. Contamination Control Services

1. **<u>Contamination Control Management Specific Tasks</u>** – The Contractor shall provide contamination control management services, including:

   a. Developing contamination control plans for spacecraft and instruments;
   b. Determining contamination control requirements and developing appropriate monitoring plans and procedures to assess contamination control requirement compliance; and
   c. Monitoring, reviewing, analyzing, and reporting on overall contamination control management, implementation, and development.

2. **Contamination Control Analysis Specific Tasks** – The Contractor shall provide contamination control analysis services, including:

    a. Developing analytical transport models (molecular and/or particulate) for spacecraft and instrument systems and/or other space flight hardware and generating contamination hazards predictions;
    b. Performing detailed environmental analyses of all phases of assembly, integration, test, transportation, pre-launch, launch, on-orbit, and descent and comparing against requirements; and
    c. Establishing surface contamination limits based on allowable performance degradation and conducting trade-off analyses, analyzing specifications, and reviewing requirements.

3. **Integration and Test Contamination Control Specific Tasks** – The Contractor shall provide integration and test contamination control services, including:

    a. Providing direction during test planning and the test preparation phase;
    b. Designing, developing, fabricating, and integrating contamination control monitoring devices;
    c. Developing procedures for specific test and cleanliness requirements; and
    d. Providing contamination control monitoring during integration and providing support during testing and integration.

4. **Contamination Laboratory Support Specific Tasks** – The Contractor shall provide contamination laboratory support services, including:

    a. Perform and record outgassing measurements in the Molecular Kinetics (MOLEKIT) facility;
    b. Provide contamination flight monitor support for fabrication, testing, integration, on-orbit data review, and data reduction; and
    c. Provide setup and operation of airborne particle counters and microscopes.

5. **Development and Use of Contamination Standards Specific Tasks** – The Contractor shall provide contamination laboratory support services, including:

    a. Analysis and support for the development of GSFC, NASA, national and international contamination control standards;
    b. Develop, procure, calibrate, and test new equipment for the purpose of developing new standards or monitoring flight projects contamination control; and
    c. Write and present papers documenting the development of new techniques and standards in contamination control.

6. **Cleaning Support and Technology Specific Tasks** – The Contractor shall provide cleaning support services, including:

   a. Provide cleaning procedures and precision cleaning, for GSE and flight hardware and
   b Investigate, develop, procure, calibrate, and test new cleaning techniques and applications to enhance our ability to provide and validate cleanliness of flight hardware.

7. **Space Environmental Coatings Testing, Applications and Management Specific Tasks** – The Contractor shall provide support services, including:

   a. Perform Flight qualification and space environment testing of coatings along with thermal radiative property measurements, thickness measurements, and coating adherence testing;
   b. Develop, operate, and maintain GSFC unique facilities to select, apply, and qualify coatings for use on spacecraft and instrument surfaces;
   c. Characterize thermal control surfaces and assess degradation from environmental effects due to UV radiation, thermal cycling, charged particles, electrostatic discharge, outgassing and humidity; and
   d. Develop and maintain a database of thermal property test data and coordinate extended shelf life testing of paints.

8. **Thin Films Specific Tasks** – The Contractor shall provide support services, including:

   a. Provide support in the area of vacuum vapor deposition and sputter deposited thin films; and
   b. Provide support to apply thin film coatings on space flight parts at either government facilities and/or the contractor facilities.

FUNCTION 3 – RESEARCH AND TECHNOLOGY SERVICES

The Contractor shall provide advanced research and technology support to EED, ISTD, SED, MESA, and related organizations. These services may include development, test and analysis work in support of the Research and Technology activities.

A. Solid State Device Research and Development

The Contractor shall provide research, design, development, and analysis of electronic devices for application to science missions.

1. **Electronic Device Development Specific Tasks** – The Contractor shall provide electronic device development services, including:

    a. Activities that support the research, development, evaluation, design, manufacture, and test of semiconductor devices in current use or in development as future technology for use in aerospace applications;
    b. Research and development of new, state-of-the-art electronic packaging techniques; and
    c. Research and implementation to flight standards of surface mount and multi-chip modules technology.

B. Instrument Systems Technology Services

The Contractor shall provide research, design, development, and testing, and analysis services for instrument systems, including:

1. **Instrument Systems Specific Tasks** – The Contractor shall provide services for the research and development of advanced analytical, engineering, integration, and testing including:

    a. System functional test techniques and methods;
    b. System performance testing techniques;
    c. Laser communications analysis;
    d. Instrument Systems signal to noise analysis;
    e. Advanced hyperspectral imaging concepts;
    f. Microwave and sub-millimeter wave radiometer advanced concepts and performance modeling;
    g. Instrument applications of computational optics; and
    h. Synthetic Aperture Radar advanced concepts;

C. Optoelectroincs Technology Services

The Contractor shall provide research, design, development, test, and analysis services for electro-optical subsystems for scientific systems, including:

1. **Optoelectronics Specific Tasks** – The Contractor shall provide services for optoelectronics technology tasks as follows:

    a. Read/write magnetoptical disk storage;
    b. Optical engineering;
    c. Opto-mechanical engineering;
    d. Fiber optic/Photonic systems;
    e. Calibration and test sets; and
    f. Alignment.

2. **<u>Opto-Mechanical Specific Tasks</u>** – The Contractor shall provide services for opto-mechanical technology tasks as follows:

   a. Develop advanced opto-mechanical technology for components and subsystems;
   b. GSE optical instrumentation;
   c. Develop novel materials for lightweight optical components, mounts, and support structures;
   d. Develop novel thin film design, fabrication, and characterization processes;
   e. State-of-the art diffractive optics and characterization;
   f. New optical design concepts and analysis techniques; and
   g. State-of-the-art optical fabrication and test methods.

D. Microwave and Millimeter Wave Technology Services

The Contractor shall provide research, design, development, test, and analysis services for microwave and millimeter wave systems for ground and space flight applications, including:

1. **<u>Antenna and Tracking Systems Specific Tasks</u>** – The Contractor shall provide services for antenna and tracking system technology tasks, including:

   a. Instrument antenna systems;
   b. Communication antenna systems;
   c. Antenna deployable/retractable assemblies;
   d. Antenna feed networks/components;
   e. Servo controls;
   f. Gimbals; and
   g. Phased array antenna technology.

2. **<u>Microwave and Millimeter Wave Components Specific Tasks</u>** – The Contractor shall provide services for microwave and millimeter wave components technology tasks, including:
   a. Research and development of high-speed analog and digital electronics for RF, microwave, and millimeter wave communications systems;
   b. Research and development of components for space communication applications, including modulator/exciters, solid-state power amplifiers, low-noise amplifiers, low-noise mixers, and improved local oscillator technology;
   c. Systems engineering; and
   d. Transponder technology.

3. **<u>Microwave Instruments Specific Tasks</u>** – The Contractor shall provide services for microwave instruments technology tasks, including:

   a. Instrument antenna systems;

b. Low noise receivers, including improved local oscillator technology;
c. Instrument antenna coverage;
d. Microwave sources;
e. Studies and models;
f. Interferometric systems;
g. Polarimetric systems;
h. Instrument system calibration; and
i. Instrument maintenance, upgrades, and field campaign support.

E. Instrument Electronics Systems Technology Services

The Contractor shall provide services for research, design, development, test, and analysis of advanced signal processing electronics for space flight systems, including support for language-based microelectronics development (VHDL, Verilog):

1. **Sensor Signal Processing Specific Tasks** – The Contractor shall provide services for sensor signal processing technology tasks, including:

   a. Systems engineering;
   b. Digital signal processing electronics development;
   c. Analog signal processing electronics development;
   d. Advanced flight hardware;
   e. Applications unique electronic test equipment;
   f. Low-level analog signal electronics development;
   g. Software engineering;
   h. Flight robotics; and
   i. Analog/Mixed-signal ASIC development.

2. **Digital Signal Processing Specific Tasks** – The Contractor shall provide services for digital signal processing technology tasks, including:

   a. Microprocessor development;
   b. Command and data handling functions (e.g, processor, command ingest, telemetry outputting, timekeeping, fault detection/correction, analog collection, custom Input/Output (I/O) interfaces, standardized data buses, data storage);
   c. Digital logic design through schematic capture and/or Hardware Description Language (HDL) coding; and
   d. Reconfigurable computing system design and implementation.

3. **Advanced Applications Specific Tasks** – The Contractor shall provide services for advanced applications technology tasks, including:

   a. Data and image compression;
   b. Data coding;

c. High-speed electronics development;
d. Packet telemetry development;
e. Reduced Instruction Set Computer (RISC) development;
f. Proof-of-concept studies; and
g. Electronic engineering design.

4. **Power Systems Specific Tasks** – The Contractor shall provide services for power systems technology tasks, including:

a. Instrument power system design and analysis; and
b. Instrument power distribution electronics.

F. Computer Support Technology Services

The Contractor shall provide computer technology services, including:

1. **Computer Support Specific Tasks** – The Contractor shall provide computer technology services, including:

a. Engineering support to analyze data acquisition, processing, distribution, archival/storage, and measurement problems;
b. Data reduction to include statistical and thematic trends analyses;
c. Diagnostics support for instrument checkout between test consoles and test components;
d. Program services to utilize test instruments in aerospace system test and analysis, including General Purpose Interface Bus (GPIB) type operation and Graphical User Interface (GUI) based software system;
e. General in-house computer software maintenance to include, but not be limited to, updating and debugging programs;
f. Design, coding, integration, test, documentation, and maintenance of special applications programs;
g. Updating of existing technical in-house computer databases;
h. Transfer of programs from one system to another and testing for functional operations and real time data transfer between dissimilar systems;
i. Debugging of general utility programs, such as graphic packages;
j. Providing support in analyzing and implementing solutions to computer hardware interface problems;
k. Providing support in network and operating system configurations, troubleshooting, installation, and maintenance; and
l. Design and debug of test procedures.
m. Management of mission test facilities that require significant computer capabilities.

2. **IT Systems Security and System Administrator Function for 560 Labs and NASA Electronic Parts and Packaging (NEPP) Program** – The Contractor shall provide Windows, Macintosh, Linux, UNIX, Web, LAN systems administration services to desktops, workstations and servers, including:

   a. Logging, reporting, diagnosing and correcting system faults;
   b. Configuring systems for performance, security and network compatibility;
   c. Performing updates of the operating system and associated software for desktops and workstations;
   d. Providing support in the preparation and updating of IT security and system administration documentation;
   e. Working with the Code 700 on implementing IT security initiatives;
   f. Providing support to users with software/hardware installation;
   g. Performing Help Desk functions including problem diagnosis and answering user questions regarding applications;
   h. Monitoring system and network security and availability;
   i. Repairing workstations, desktops and printers on an emergency basis; and
   j. Data backup, archive, and retrieval.

   In addition to any other requirements of this contract, all individuals who perform tasks as a system administrator or have authority to perform tasks normally performed by system administrator shall be required to demonstrate knowledge appropriate to those tasks. This demonstration, referred to as the NASA System Administrator Security Certification, is a NASA funded two-tier assessment to verify that system administrators are able to –

   1. Demonstrate knowledge in system administration for the operating systems for which they have responsibility; and
   2. Demonstrate knowledge in the understanding and application of Network and Internet Security.

   Certification is granted upon achieving a score above the certification level on both an Operating System Test and the Network and Internet Security Test. The Certification earned under this process will be valid for three years. The criteria for this skills assessment has been established by the NASA Chief Information Officer. The objectives and procedures for this certification can be obtained by contacting the IT Security Awareness and Training Center at (216) 433-2063.

   A system administrator is one who provides IT services, network services, files storage, web services, etc. to someone else other than themselves and takes or assumes the responsibility for the security and administrative controls of that service or machine. A lead system administrator has responsibility for information technology security (ITS) for multiple computers or network devises represented within a system; ensuring all devices assigned to them are kept in a secure configuration (patched/mitigated); and ensuring that all other system administrators under their lead understand and perform ITS duties. An individual that has full access or arbitrative rights on a system or machine that is only servicing him or herself does not constitute a "system administrator" since that individual is only providing or accepting responsibility for their system. Such an individual that is only is not required to obtain a System Administrator Certification.

1. **Web Page Development and Maintenance Function** – The Contractor shall provide web development services to help promote organizational capabilities, including:

   a. Development, maintenance, and upgrade of web sites;
   b. Compliance to Agency and Center policy (GSFC Webmaster) such as 508 compliance and Post 9-11 accessibility compatibility;
   c. Defining with customer the look and feel of the web site, and reviewing web site requirements;
   d. Developing prototype web sites for maturing web based concepts; and
   e. Providing maintenance services to keep web site up to date and 508 compliant.

G. Thermal Control Systems Technology Services

The Contractor shall provide technology services for the research and development of advanced, two-phase thermal control systems, and cryogenic systems for instruments, systems, and spacecraft, including:

1. **Thermal Control Systems Specific Tasks** – The Contractor shall provide thermal design, fabrication, test, and analysis technology services for thermal control systems, including:

   a. Engineering design, analysis, fabrication, testing, and analysis for fixed and variable conductance heat pipes, diode heat pipes, and phase change materials;
   b. Design, analysis, fabrication, and test of mechanically and capillary pumped thermal control systems;
   c. Systems engineering to define and recommend appropriate enhancements to existing thermal control technology;
   d. Testing facilities and services for testing two-phase heat transfer systems containing ammonia and other working fluids;
   e. Thermal/mechanical capabilities to perform Phase A and Phase B studies for proposed flight experiments to demonstrate two phase heat transfer technology;
   f. Thermal/mechanical/electrical/system engineering capabilities to perform testing on mechanical cryocoolers. This includes the design, fabrication, and testing of cryocooler drive electronics and analysis of thermal performance test results;
   g. Mechanical/electrical/thermal support for on-going flight projects using mechanical cryocoolers;
   h. Variable emittance thermal control surfaces, alternative materials; and
   i. Supporting the development of thermal flight experiments.

H. Power Systems Technology Services

The Contractor shall provide research, design, development, evaluation, and qualification of power system designs and components for space flight applications, including:

1. **<u>Power System Architecture Specific Tasks</u>** – The Contractor shall provide services for power system architecture technology, including:

   a. Direct energy transfer; and
   b. Peak power tracker.

2. **<u>Energy Conversion Devices Specific Tasks</u>** – The Contractor shall provide services for energy conversion device technology, including:

   a. Multi-junction solar cells;
   b. Thin film solar cells; and
   c. Concentrators.

3. **<u>Energy Storage Devices Specific Tasks</u>** – The Contractor shall provide services for energy storage device technology, including:

   a. Li Ion;
   b. Flywheel technology;
   c. Fuel cells; and
   d. Li Polymer.

4. **<u>Power Electronics Specific Tasks</u>** – The Contractor shall provide services for power electronics technology, including:

   a. Low and High voltage dc-dc converters; and
   b. Microprocessor controlled spacecraft power management and distribution.

I. Data Systems Technology Services

The Contractor shall provide technology services formulating and developing advanced technology concepts and the advancement of its associated technology readiness level (TRL).

1. **<u>Data Processing Technology Specific Tasks</u>** – The Contractor shall provide services, including:

   a. Plan and accomplish the infusion of the resulting technology into mission or project systems for data processing or the implementation of data processing prototypes; and

b. Design and develop advanced concepts for all aspects of mission including hardware, complex electronics, reconfigurable computing, and data solutions to meet highly complex mission data processing objectives.

J. Demonstration, Presentation and Conference Services

The Contractor shall provide technology services for hardware and software demonstrations, technical/project/conference presentations, and conference planning/implementation for items within the scope of this Statement of Work, including:

1. **Demonstration Specific Tasks** – The Contractor shall provide hardware, software, support equipment, and technical services for onsite and offsite demonstrations.

2. **Presentation Specific Tasks** – The Contractor shall provide materials for inclusion in technical/project/conference presentations, including viewgraphs, information, photographs, etc. In addition, the Contractor shall perform the presentation.

3. **Conference Specific Tasks** – The Contractor shall provide support to the Government by providing services in the planning and implementation of conferences.

K. Guidance, Navigation and Control (GN&C) Technology Services

The Contractor shall provide services for the research and development of advanced GN&C technology for systems, subsystems, components, devices, and elements for spacecraft, balloons, UAV's, sounding rockets, instruments, systems, and other platforms.

1. **Advanced GN&C Sensor/Actuator Design** – The Contractor shall provide hardware systems engineering to design, develop and test sensor/actuator hardware and software.

K. Cryogenic and Fluid Systems Technology Services

The Contractor shall provide technology services for developing advanced cryogenic and fluid systems to support new NASA missions and applications:

1. **Cryogenic and Fluids Systems Specific Tasks** – The Contractor shall provide services, including:

   a. Structural and thermal interfaces to mechanical refrigerators;
   b. Components for an advanced adiabatic demagnetization refrigerator;
   c. Cryogenic actuators; and

d. Design & develop monopropellant, bi-propellant and cryogenic fluid transfer systems for in-space refueling.

M. Advanced Coating and Film Technology Services

The Contractor shall provide technology services for developing advanced coating and film technologies to support new NASA missions and applications:

1. **Advanced Coating and Film Technology Specific Tasks** – The Contractor shall provide services, including:

    a. Developing, procuring, calibrating, testing, and maintaining new or existing coatings and films; and
    b. Writing and presenting papers documenting the development and application of this new technology.

FUNCTION 4 – PARTS AND MATERIALS PROGRAM SERVICES

The Contractor shall provide parts and materials services for hardware/software within the scope of this contract, including:

A. Materials and Processes Services

1. **Materials and Processes Specific Tasks** – The Contractor shall provide materials and process services, including:

    a. Contamination analysis and control;
    b. Construction analysis;
    c. Design compatibility; and
    d. Material evaluation,

B. Protective Coating and Encapsulation Services

1. **Protective Coating and Encapsulation Specific Tasks** – The Contractor shall provide protective coating and encapsulation services, including:

    a. Iridite, anodize, or use comparable coating processes to finish metal surfaces;
    b. Conformal coat and/or encapsulate components, parts, and fixtures in accordance with task orders and applicable flight application documentation; and
    c. Prime and paint surfaces, parts, and assemblies

C. Component and Parts Labeling Services

1. **<u>Component and Parts Labeling Specific Tasks</u>** – The Contractor shall provide component and parts labeling, including:

   a. Label and identify parts with location as specified by the applicable document or drawing; and
   b. Cover labels with GSFC approved protective finish where required.

D. Electronics Packaging Services

The Contractor shall provide Space Electronics Packaging, Manufacturing & Assembly Services, including:

1. **<u>Packaging Design</u>**

   a. Circuit board design and layout
   b. Enclosure design and layout
   c. Mechanical mounting including thermal and vibration isolation
   d. EMI/EMC isolation and protection
   e. Ground support equipment including mechanical electrical and optical
   f. Mechanical, Thermal, Contamination analysis
   g. Radiation mitigation analysis

2. **<u>Manufacturing and Assembly Services</u>**

   h. Circuit board procurement, manufacturing, and assembly
   i. Enclosure and circuit board mechanical parts manufacturing and assembly
   j. Manufacturing and assembly of optical, mechanical, detector, electrical/electronics, RF and microwave boards and components
   k. Test equipment and test fixture manufacturing and assembly
   l. Electrical test assembly support
   m. Wiring harness manufacturing and assembly
   n. Mechanical and electronics protective coatings
   o. Implementation of advanced technology manufacturing and assembly services/techniques, such as Column Grid Arrays, Chip-On-Board, etc.
   p. Ground support equipment manufacturing and assembly, including mechanical, electrical, and optical

E. Electrical Electronic Electromechanical (EEE) Parts

The Contractor shall provide parts engineering support for GSFC and other NASA Centers. Support for operations of GSFC's Board Layout Facility and Assembly Facility included all the functions identified in the two previous sections (4.D.1 and 4.D.2).

1. EEE Parts Services

A. **<u>Project Parts Engineering and Program Management Specific Tasks</u>** – The Contractor shall provide parts engineering and program management services, including:

a. Commodity Expertise for all EEE part categories listed in EEE-INST-002, and Detectors, Microwave, and Fiber optics/Photonics;
b. Establishment of parts selection and approval criteria for Grade 1, 2 and 3 level programs and projects;
c. Develop, review and implement Parts Control Plans;
d. Interface with Designers, Radiation Engineers and Chief Safety and Mission Assurance Officer (CSO) in parts selection and approval process;
e. Develop, review and maintain parts lists (PIL, PAPL, ADPL, & ABPL);
f. Conduct and participate in Parts Control Boards;
g. Review and disposition all Government-Industry Data Exchange Program (GIDEP) Alerts and NASA Advisories;
h. Specification preparation and interfacing with manufacturers for custom flight parts;
i. Investigations of reliability issues related to parts failures;
j. Evaluation of advanced technology parts;
k. Prepare and implement a plan to address issues related to EEE Parts with lead free finishes; and
l. Prepare and implement a plan to address issues related to the assembly of Printed Wiring Boards using lead free solders.

B. **<u>EEE Parts Procurement, Storage, and Kitting</u>**

a. Procure critical EEE parts and supporting materials for Engineering and Flight Hardware Development at Goddard.
b. Perform expediting services for schedule critical purchase orders and kit requests.
c. Perform receiving inspection of EEE Parts upon delivery, to include kind, count, condition, and X-ray fluorescence (XRF) inspection, etc.

d. Provide storage and inventory management for EEE Parts at the contractor facility equivalent to 2000 square feet.
e. Perform kitting of EEE Parts upon request by project parts engineers or flight projects. Provide tools, materials, and resources required for kitting efforts such as counting, dividing part lots, repackaging parts, updating inventories, and observing ESD precautions.
f. Provide weekly reporting and tracking of purchase requests, outstanding orders, expected delivery dates, receiving inspection, inventory status, and kitting.
g. Provide a software tool to implement these functions accurately and efficiently.

C. **Parts Testing and Analysis Laboratory Specific Tasks** – The Contractor shall provide parts testing and analysis services, including:

a. Failure Analysis – Conduct analysis of failed parts, materials and components to determine and classify the physical mechanisms of failures.
b. Destructive Physical Analysis (DPA) – Conduct DPA to determine if the part quality and workmanship meet the requirements of applicable NASA or Manufacturer specification. DPA shall include electrical, mechanical, environmental and analytical testing.
c. Incoming Test and Inspection, Screening, Evaluation and qualification testing - Conduct these tests in accordance with appropriate part specification and test procedures. Analyze test results and provide summary reports and certification logs as required.
d. Test Equipment specific tasks – Support the calibration, maintenance and upgrading of test equipment related activities and support property system requirements.
e. Ensure safe operating procedures, practices and availability of personal protection equipment in the laboratory.

D. **Photonics Tasks** - The Contractor shall provide following manufacturing and test services:

a. Building fiber optic cables and arrays for spaceflight and ground test using a wide variety of fiber optic components.
b. Conducting investigations into fiber optic and photonic reliability, radiation effects and other related engineering tasks, including design of optical fiber and optical free space components;
c. Operating a Technology Validation laboratory with a variety of environmental, optical microscopy and x-ray microscopy equipment for the purpose of investigating the reliability of EEE parts and technologies; and
d. Evaluating new photonics technologies for suitability for use in space flight application.

e. Integration and test services of flight and engineering model photonic components and systems including fiber optic cables, connectors, and feed throughs.

E. **<u>Parts Information System Specific Tasks</u>** – The Contractor shall provide parts information system services, including:

a. Develop, update, and maintain an electronic parts database, that includes candidate parts lists, as designed parts lists and approved parts lists of all GSFC projects;
b. Develop, update, and maintain the on-line NASA Parts Selection List (NPSL), and EEE Parts Selection, Qualification, Screening and Derating Guidelines (EEE-INST-002);
c. Update and maintain Parts Analysis Lab website, NASA Electronics Parts & Packaging (NEPP) website and the Parts Engineers website;
d. Maintenance of the Parts Analysis (PA) laboratory network to include on line storage of all test reports; and
e. Maintenance of EEE parts specification library. This includes working with the NASA Standard's organization.

FUNCTION 5 – DOCUMENTATION SERVICES

The Contractor shall provide documentation services for all levels of hardware and software within the scope of this Statement of Work, as specified in task orders. Documents shall conform to applicable documents and specifications. These shall include pertinent NASA Handbook (NHBs), Safety and Mission Assurance Programs (SMAP), quality standards, GSFC standards, documents of other NASA Centers, Federal standards, military standards, and commercial standards.

A. Document Services

The Contractor shall provide documentation services, including instrument conceptual designs, program plans, systems analyses, illustrations, technical and implementation plans, test plans, test procedures, test scripts, software documentation, and the full range of system hardware and software documentation. These shall also include up-to-date drawings, specifications, certifications, reports, interface control documents, and agreements.

1. **<u>Document Services Specific Tasks</u>** – The Contractor shall provide electronic media and document services, including:

a. Technical writing;
b. Editing;
c. Drafting;
d. CAD/CAM;

e. Photographic;
f. Video;
g. Reproduction;
h. CD and DVD; and
i. Posters and Displays.

2. **Photo and Video Specific Tasks** – The Contractor shall use photos and video for maintenance, engineering, or as documentation to explain a problem. They shall become supplemental to assist in unit repair or future development and maintenance. A scale shall be included to indicate relative dimensions in photographs and/or video, where appropriate.

FUNCTION 6 – MAINTENANCE SERVICES

The Contractor shall provide maintenance support, including:

A. Preventive Maintenance

The Contractor shall perform preventative maintenance on hardware and software within the scope of this Statement of Work as specified in task orders.

B. Emergency Repair Services

The Contractor shall provide expeditious emergency repair services for hardware and software within the scope of this Statement of Work, as specified in task orders. The Contractor shall respond to the Government within four hours of notification to determine and implement a mutually agreeable course of action. In some cases, there shall be 24-hour coverage during flight hardware and software evaluation, verification, and test. This service shall comprise of repair, modification, or replacement of components, codes, subassemblies, and assemblies. Documentation updates shall be required as a result of any design change.

FUNCTION 7 – SUSTAINING ENGINEERING SERVICES

The Contractor shall provide sustaining engineering services for hardware and software within the scope of this Statement of Work, including:

1. Modifications of hardware/firmware, including installation of circuits for improved reliability and/or performance;

2. Modifications of wiring to improve circuit performance;

3. Fabrication, assembly, wiring, and testing of printed circuit assemblies where necessary to update old circuitry or improve reliability;

4. Engineering, fabrication, testing of assemblies or sub-assemblies to replace outdated circuitry to eliminate component or circuit failures;

5. Engineering, fabrication, assembly, and testing of engineering circuits to correct problems encountered;

6. Modifications of mechanical assemblies, structures, and mechanisms to correct or improve the design; and

7. Update of drawings, manuals, and technical data to reflect current status at the time of modifications.

FUNCTION 8 – EDUCATION SERVICES

The Contractor shall provide support and/or development of education services. This support encompasses a wide range of activities that include but are not limited to: a) the development of engineering/scientific educational courses either by providing support to NASA/Goddard in putting together educational topics or completely developing a topic based on the Contractor's area of expertise; b) the development of partnerships with universities for the purpose of educating the next generation engineers and scientists on state-of-the-art spacecraft and instrument systems; c) the development of specific engineering topics, such as Advancement in Spacecraft Avionics Systems; and d) the development of Programs, such as internships that help feed the pipeline for NASA's future engineers and scientists. Some of the specific areas the Contractor may be asked to support are the following:

1. AETD Engineering mini-course series;

2. GSFC's partnership with the University of Maryland Aerospace Department to teach courses in satellite design;

3. The NASA Engineering Training (NET) activity; and

4. Programs with universities and NASA headquarters.

FUNCTION 9 – STANDARDS AND PROCESS

The Contractor shall provide support for engineering standards work and engineering process work, including:

1. ISO documentation and process generation;

2. Engineering standards documentation and review;

3. Engineering process documentation; and

4. Activities in support of engineering process improvement.

APPLICABLE DOCUMENTS AND SPECIFICATIONS

The Contractor shall adhere to all applicable portions of the following documents and specifications in the performance of this contract. Most all documents can be found at NASA's Technical Standards Program (https://standards.nasa.gov); NASA's Online Directives Information System (NODIS) (http://nodis3.gsfc.nasa.gov); the Goddard - Directives Management System (GDMS) (http://gdms.gsfc.nasa.gov). The latest updated version shall apply:

General:
NPR 7120.5, "NASA Space Flight Program and Project Management Requirements"
NASA SP-2007-6105 Rev1, The NASA Systems Engineering Handbook -
NPR-7150.2, "NASA Software Engineering Requirements"
NASA-STD-8719.13, "NASA Software Safety Standard"
GPR 7123.1 "Systems Engineering"
GPR 8070.4, "Administration and Application of the Goddard Rules for the Design, Development, Verification and Operation of Flight Systems"
GSFC-STD-1000, "Rules for the Design, Development, Verification, and Operation of Flight Systems."

Safety and Mission Assurance:
NSTS 1700.7 (Addendum) "Safety Policy and Requirements for Payloads Using the International Space Station".
NASA-STD-8719.13, "NASA Software Safety Standard"
NASA-STD-8719.24 (Base), "NASA Expendable Launch Vehicle Payload Safety Requirements"
NASA-STD-8719.24 (Annex), "Annex to NASA-STD-8719.24 NASA Expendable Launch Vehicle Payload Safety Requirements",
GPR 1700.1A, "Goddard Space Flight Center Occupational Safety Program" 500-PG-8715, "AETD Safety Manual"
AS9100C, "Quality Management System Standard"
ISO 9001, "Quality Management"
GPR 8730.1, "Metrology, Control of Measuring and Test Equipment"

EEE Parts
EEE-INST-002, "Instructions for Selection, Screening, and Qualification of Electrical, Electronic, and Electromechanical (EEE) Parts"
500-PG-4520.2.1, "Electrical, Electronic, and Electromagnetic Counterfeit Parts Avoidance Plan"

Conformal Coating and Staking:
NASA-STD-8739.1, "Workmanship Standard for Polymeric Application on Electronic Assemblies"

Soldering – Flight, Surface Mount Technology:
NASA-STD-8739.2, "Workmanship Standard for Surface Mount Technology"

Soldering – Flight, Manual (hand):
NASA-STD-8739.3, "Soldered Electrical Connections"

Soldering – Ground Systems:
Association Connecting Electronics Industries (IPC)/Electronics Industry Alliance (EIA) J-STD-001D AMD1, "Requirements for Soldered Electrical and Electronic Assemblies"

Electronic Assemblies – Ground Systems:
IPC-A-610, "Acceptability of Electronic Assemblies"

Crimping, Wiring, and Harnessing:
NASA-STD-8739.4, "Crimping, Interconnecting Cables, Harnesses, and Wiring" 565-PG-8700.2.1 "Design and Development Guidelines for Spaceflight Electrical Harnesses"

Fiber Optics:
NASA-STD-8739.5 "Fiber Optic Terminations, Cable Assemblies, and Installation"

Electro-Static Discharge (ESD) Control:
GPR 8730.6, Electrostatic Discharge (ESD) Control

Printed Wiring Board (PWB) Design:
500-PG-8700.2.2, Electronics Design and Development Guidelines 500-PG-8700.2.4, Mechanical Design and Development Guidelines,
GSFC X-673-64-1F, "Engineering Drawing Standards Manual" (December 1994) IPC-2221, "Generic Standard on Printed Wiring Board Design"
IPC-2222, "Sectional Standard on Rigid Printed Wiring Boards Design" IPC-2223A, "Sectional Design Standard for Flexible Printed Boards"

PWB Manufacture:
IPC A-600, "Acceptability of Printed Boards"
IPC-6011, "Generic Performance Specification for Printed Boards"
IPC-6012, "Qualification and Performance Specification for Rigid Printed Boards" Flight Applications – Supplemented with: GSFC/S312-P-003, Procurement Specification for Rigid Printed Boards for Space Applications and Other High Reliability Uses
IPC-6013 "Qualification and Performance Specification for Flexible Printed Boards" IPC-6018 "Microwave End Product Board Inspection and Test"

Section 508 Electronic and Information Technology (EIT) Standards (http://www.access-board.gov/sec508/standards.htm).

1194.21 Software Applications and Operating Systems
1194.22 Web-based Intranet and Internet Information and Applications
1194.24 Video and Multimedia Products

Mechanical Design
541-PG-8072.1.2 “GSFC Fastener Integrity Requirement”

Testing
GSFC-STD-7000 “General Environmental Verification Standard (GEVS)

REFERENCE DOCUMENTS AND SPECIFICATIONS

The following documents and/or specifications are provided as reference material for the performance of this contract. The last updated version shall apply:

NPR 7120.5, “NASA Space Flight Program and Project Management Requirements”*
NPR 7120.5, “NASA Space Flight Program and Project Management Handbook, Feb 2010”
SP-6105, “The NASA Systems Engineering Handbook, December 2007”
* Note: NASA Interim Directive (NID) NM 7120-97 in effect until revised to version “E”